

IINTS-AF Jury Physiology Brief

Competition evidence bundle for EUCYS review

Purpose: A short, judge-friendly answer to the question: "What is the SDK actually simulating?"

This PDF was rendered from the markdown report and packaged benchmark artifacts inside the IINTS-AF SDK workspace.

The 30-Second Answer

IINTS-AF simulates virtual-patient glucose physiology, not just lines on a graph. It represents meals, insulin action, insulin-on-board, carbohydrates-on-board, exercise, sensor imperfections, and safety supervision as separate pieces so they can be tested and explained.

The Three Numbers Judges Should Remember

Number	Meaning
70-180 mg/dL	standard glucose target range used for Time in Range
<70 mg/dL	low-glucose exposure
<54 mg/dL	clinically more serious hypoglycemia threshold

These values make a simulated run interpretable, but they do not turn a simulator into a clinical device.

The Three Layers Judges Should Not Confuse

Layer	Example
physiology	true glucose rises after a meal
measurement	the CGM signal is noisy or delayed
protection	the supervisor blocks an excessive insulin request

This is one of the SDK's main strengths: it does not hide those ideas inside one black-box score.

One Meaningful Example

Event	Example
breakfast	48 g carbs
lunch	62 g carbs
exercise	added glucose-lowering pressure
dinner	74 g carbs

A believable glucose graph should react to events like these. A perfectly smooth curve can be easier to code and less realistic scientifically.

Why Data Quality Matters

Bad CGM data can quietly mislead an algorithm. Missing insulin, duplicated timestamps, impossible glucose jumps, and sensor artifacts can make a model look better or worse for the wrong reason. That is why IINTS-AF includes MDMP data checks before evidence is trusted.

What The SDK Does Not Claim

- it is not a medical device
- it does not dose real patients
- it is not a personalized digital twin
- simulation does not replace clinical evidence

Best One-Sentence Claim

IINTS-AF makes insulin-algorithm research more reproducible by keeping physiology, data quality, and safety behavior visible instead of hiding them inside one polished demo graph.